ZERTIFIKAT ◆ CERTIFICATE ◆ 認証書 ◆ СЕРТИФИКАТ ◆ CERTIFICADO ◆ CERTIFICAT

Product Service

EC-Attestation of Compliance

No. E8N 07 12 46532 012

Holder of Certificate: **AJINEXTEK CO., LTD.**
9-3, Holim-dong, dalseo-gu
Daegu-city 704-240
REPUBLIC OF KOREA

Name of Object: **Information Technology Equipment**
Motion Control/Digital IO Board Set

Model(s): **BPFR/BPHR/SIO-DB32P/SMC-2V03/**
C6868/C6836/T68-PR/T36-PR

Description of Object:

Rated input voltage:	230 V a.c.
Rated input frequency:	50 Hz

Tested according to:
EN 55022:1998+A1:2000+A2:2003 Class A
EN 55024:1998+A1:2001+A2:2003
EN 61000-3-2:2000+A2:2005
EN 61000-3-3:1995+A1:2001+A2:2005

This EC-Attestation of Compliance is issued according to the Directive 2004/108/EC relating to electromagnetic compatibility on a voluntary basis. It confirms that the listed apparatus complies with all essential requirements of the EMC directive and applies only to the sample and its technical documentation submitted to TÜV SÜD Product Service GmbH for testing and certification. See also notes overleaf.

Test report no.: EMC-CE-1220

Date, 2008-01-07

216416

CE After preparation of the necessary technical documentation as well as the conformity declaration the required CE marking can be affixed on the product. Other relevant directives have to be observed.

TÜV SÜD Product Service GmbH · Zertifizierstelle · Ridlerstrasse 65 · 80339 München · Germany

EU Declaration of Conformity (EMC)

Type of equipment:	**Motion Control/Digital IO Board Set**
Model Name:	**BPFR/BPHR/SIO-DB32P/SMC-2V03/ C6868/C6836/T68-PR/T36-PR**
Applicant:	**AJINEXTEK CO.,LTD.**
Address:	**9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea**
Manufacturer:	**AJINEXTEK CO.,LTD.**
Address:	**9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea**

We hereby declare, that all major safety requirements, concerning to CE Mark Directive(93/68/EEC) Electromagnetic Compatibility Directives(2004/108/EC,92/31/EEC) are fulfilled, as laid out in the guideline set down by the member states of the EEC Commission. This declaration is valid for all samples that are part of this declaration, which are manufactured according to the production charts appendix.

The standards relevant for the evaluation of electrical safety requirements are as follows :

EMC : EN 55022:1998+A1+A2:2003, EN 55024:1998+A1+A2:2003
EN 61000-3-2:2000+A2:2005, EN 61000-3-3:1995+A1+A2:2005

1. Certificate of conformity / Test report issued by :

EMC : **EMC Compliance Laboratory**
Certificate No. : **EMC-CE-1220**

2. Technical documentation kept at :

AJINEXTEK CO.,LTD.
which will be made available upon request.

AJINEXTEK CO.,LTD.
9-3 Horim-dong, Dalseo-gu,
Daegu 704-240, Korea

(place and date of issue)

(name and signature of authorized person)

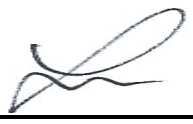
EMC TEST REPORT

Test report no.:	**EMC-CE-1220**
Type of equipment:	**Motion Control/Digital IO Board Set**
Model Name:	**BPFR/BPHR/SIO-DB32P/SMC-2V03/ C6868/C6836/T68-PR/T36-PR**
Applicant:	**AJINEXTEK CO.,LTD.**
Manufacturer:	**AJINEXTEK CO.,LTD.**
Test standards:	**EN 55022:1998+A1:2000+A2:2003, CLASS A**
	EN 55024:1998+A1:2001+A2:2003
	EN 61000-3-2:2000+A2:2005
	EN 61000-3-3:1995+A1+A2:2005
Testing Laboratory:	**EMC Compliance Ltd.**
Test result:	**Complied**

This product complies with the requirements of the EMC Directive 2004/108/EC.
The results in this report apply only to the sample tested.
This test report shall not be reproduced except in full, without the written Approval of EMC compliance Laboratory.

Date of test: 2007. 11.01 ~ 11.08

Date of Issue: 2007. 11. 13

Tested by:  KIM, JIN-SEOK

Approved by:  CHUNG, MIN-SEOK

EMC Compliance Ltd.
82-1 Jeil-ri, Yangji-myun, Churingu, Yongin-city, Kyunggi-do 449-825, Korea
TEL: 82 31 336 9919 FAX : 82 31 336 4767
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1. Client information

Applicant:	**AJINEXTEK CO.,LTD.**
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone number:	+82-53-593-3700
Facsimile number:	+82-53-593-3703

Manufacturer:	**AJINEXTEK CO.,LTD.**
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone number:	+82-53-593-3700
Facsimile number:	+82-53-593-3703

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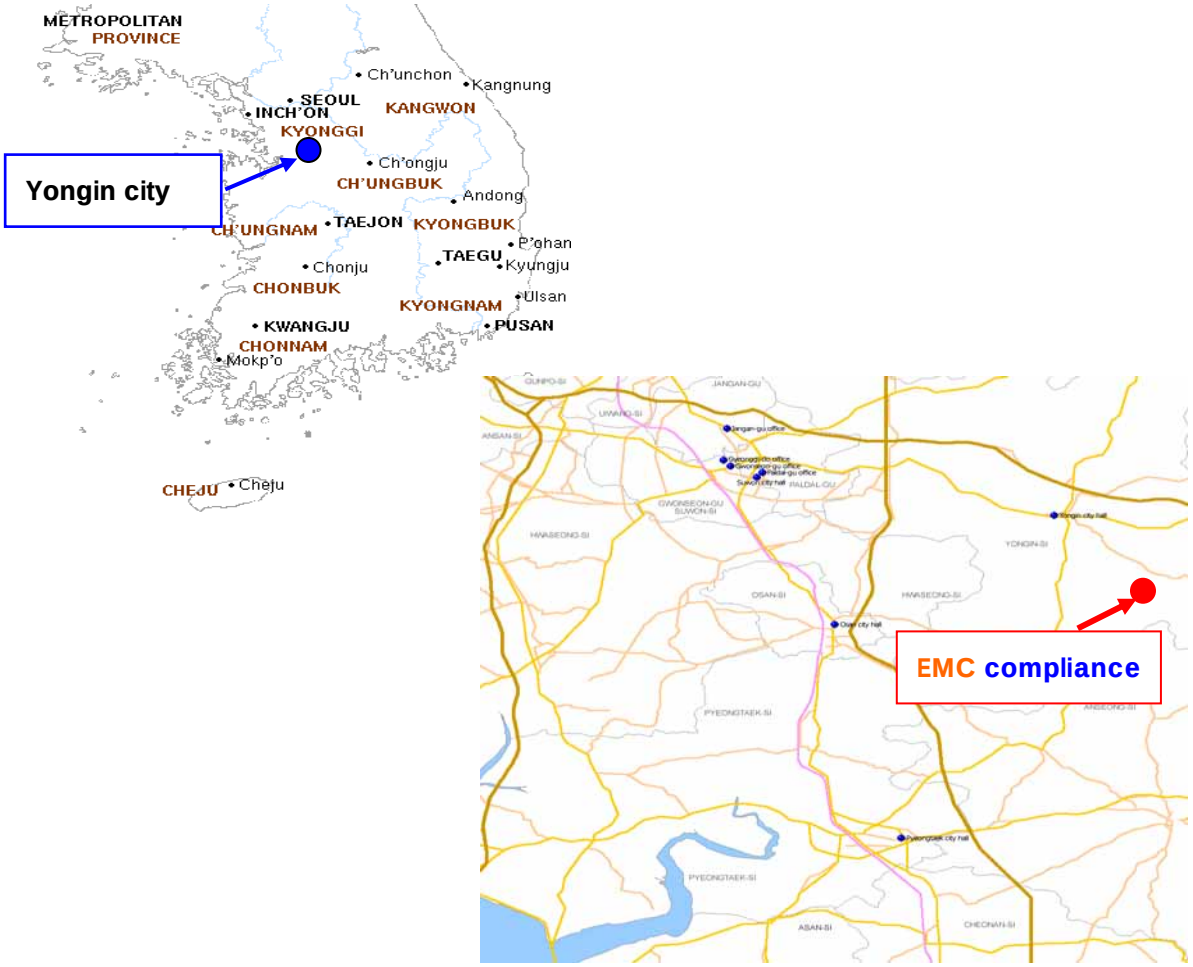
2. Laboratory information

Address

EMC compliance Ltd.
82-1 Jeil-ri, Yangji-myun, Churingu, Yongin-city, Kyunggi-do 449-825, Korea
Telephone Number : 82 31 336 9919
Facsimile Number : 82 31 336 4767

CBTL Testing Laboratory
FCC Filing No.: 793334
FCC CAB.: KR0040
VCCI Registration No. : C-1713, R-1606, T-258
KOLAS NO.: 231

SITE MAP



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3. Test system configuration

3.1 Operation environment

	Temperature	Humidity	Pressure
OATS :	16 °C	39 %	-
Shielded room :	16 °C	42 %	-
Immunity area :	16 °C	42 %	1001 hPa

Test site

These testing items were performed following locations;

Shielded Room : Conducted Emission
OATS (10m) : Radiated Emission
Immunity area : ESD, RS, EFT/Burst, Surge, CS, Dip/Interruption, Harmonics, Flicker

3.2 Measurement Uncertainty

All measurements involve certain levels of uncertainties, especially in field of EMC. The factors contributing to uncertainties are test receiver, cable loss, antenna factor calibration, Antenna directivity, antenna factor variation with height, antenna phase center variation, antenna frequency interpolation, measurement distance variation, site imperfection, mismatch, and system repeatability. Based on NIS 80, 81, the measurement uncertainty level with a 95% confidence level was applied.

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3.3 Sample calculation

Conducted Emission

The field strength is calculated by adding the LISN factor, cable loss from the measured reading.
The sample calculation is as follows :
FS = MR + LF + CL
MR = Meter Reading
LF = LISN Factor
CL = Cable Loss
If MR is 30dB, LISN Factor 1dB, CL 1dB
The result (MR) is
30 + 1 + 1 = 32dBuV

Radiated emission

The field strength is calculated adding the antenna Factor, cable loss and, Antenna pad adding, subtracting the amplifier gain from the measured reading.
The sample calculation is as follows :
FS = MR + AF + CL + AT -AG
MR = Meter Reading
AF = Antenna Factor
CL = Cable Loss
AP = Antenna Pad
AG=Amplifier Gain
If MR is 30dB, AF 12dB, CL 5dB, AP 10dB, AG 35dB
The result (MR) is
30 + 12 + 5 + 10 –35 = 22dBuV/m

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4. Description of EUT

4.1 Product description

Applicant:	AJINEXTEK CO.,LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Manufacturer:	AJINEXTEK CO.,LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Type of equipment:	Motion Control/Digital IO Board Set
Basic Model:	BPFR/BPHR/SIO-DB32P/SMC-2V03/ C6868/C6836/T68-PR/T36-PR
Serial number:	-
Testing voltage	230Vac, 50Hz
Rating:	230Vac, 50Hz

4.2 Peripherals

Description	Model / Part #	Serial number	Manufacturer
PC	-	-	ADVANTECH
PS/2 keyboard/mouse	SPR-8695I	6CEZ052658	SEJIN ELECTORN
T68-PR	T68-PR	-	AJINEXTEK
T36-PR	T36-PR	-	AJINEXTEK
Monitor	52S-S	N379HVEY216557V	SAMSUNG

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4.3 Operating conditions

- Enterprise offer program test.

4.4 Used cables

- BPFR

Start		END		Cable Spec.	
Name	**I/O Port**	**Name**	**I/O Port**	**Length**	**Shield**
EUT	C6836-2TS cable(1~4)	T-36-PR	C6836-2TS cable(1~4)	2.0	Shield
	C6868-2TS cable(1~4)	T-68-PR	C6868-2TS cable(1~4)	2.0	Shield

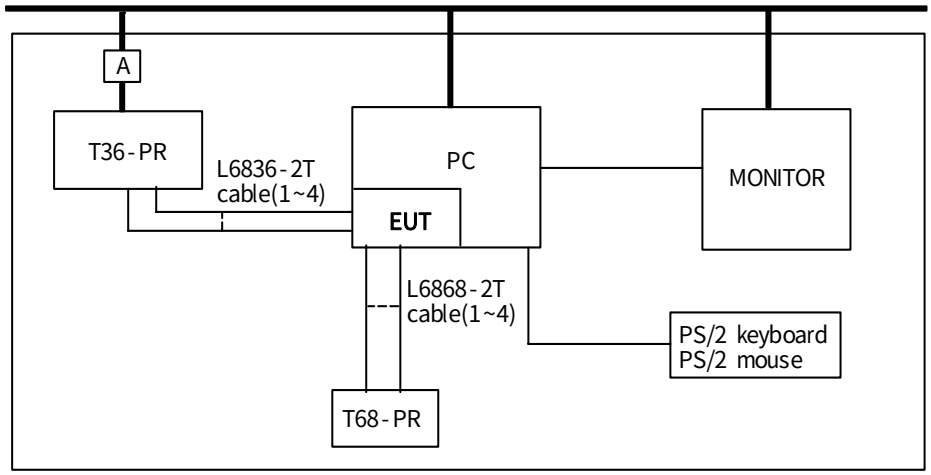
- BPHR

Start		END		Cable Spec.	
Name	**I/O Port**	**Name**	**I/O Port**	**Length**	**Shield**
EUT	C6836-2TS cable(1~2)	T-36-PR	C6836-2TS cable(1~2)	2.0	Shield
	C6868-2TS cable(1~2)	T-68-PR	C6868-2TS cable(1~2)	2.0	Shield

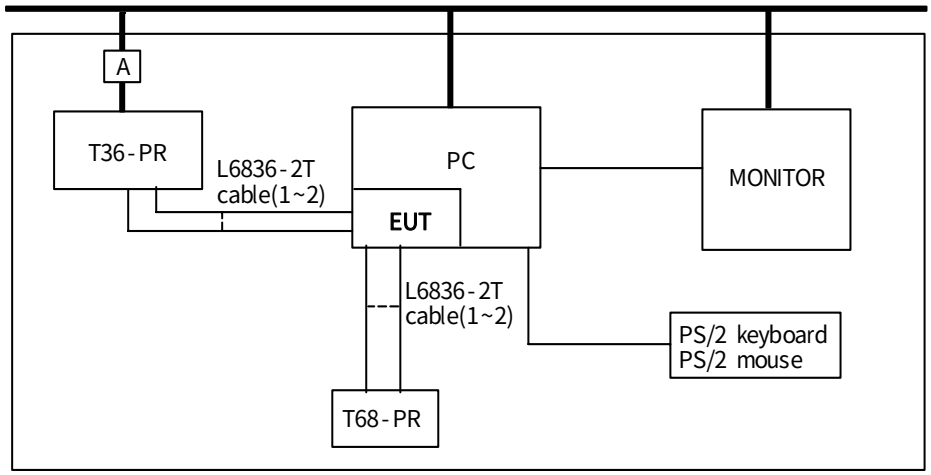
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4.5 E.U.T. test configuration

- BPFR



- BPHR



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5. Summary of test results

5.1 Modification to the E.U.T.

None

5.2 Standards & results

The following standards have been applied:

EN 55022:1998+A1:2000+A2:2003

Information technology equipment – Radio disturbance characteristics - Limits and methods of measurement

Test items	Result
Conducted emission	Pass
Radiated emission	Pass

EN 55024:1998+A1:2001+A2:2003

Information technology Equipment – Immunity characteristics - Limits and Methods of measurement

Test items	Test methods	Result
Electrostatic discharge	EN 61000-4-2:1995+A1:1998+A2:2001	Pass
Electromagnetic field	EN 61000-4-3:1995+A1:1998+A2:2001	Pass
Electric fast transients	EN 61000-4-4:1995+A1:2001+A2:2004	Pass
Surge	EN 61000-4-5:1995+A1:2001	Pass
Conducted Immunity	EN 61000-4-6:1995+A1:2001	Pass
Voltage dip/interruption	EN 61000-4-11:1994+A1:2001	Pass

EN 61000-3-2:2000+A2:2005

Limits for harmonic-current emissions (equipment input current up to including 16A per phase)

Test items	Test methods	Result
Harmonics	EN 61000-3-2:2000+A2:2005	Pass

EN 61000-3-3:1995+A1:2001+A2:2005

Limitation of voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16A per phase and not subject to conditional connection

Test items	Test methods	Result
Flicker	EN 61000-3-3: 1995+A1:2001+A2:2005	Pass

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5.3 Performance criterion

Performance criterion A: The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion B: After the test, The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test.
If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as untended.

Performance criterion C: Loss of function is allowed, provided the function is self-recoverable or can be restored by the operating of the controls by the user In accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

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6. Test results

6.1 Conducted Emission

6.1.1 Measurement procedure

Mains

The measurements were performed in a shielded room. EUT was placed on a non-metallic table height of 0.8 m above the reference ground plane. The rear of table was located 0.4 m to the vertical conducted plane. EUT was power through the LISN, which was bonded to the ground plane. The LISN power was filtered. Each EUT power lead, except ground (safety) lead, was individually connected through a LISN to input power source. All I.O cables are positioned to simulate typical actual usage according to the test standard. Both lines of power cord, hot and neutral, were measured.

6.1.2 Used equipments

Equipment	Model	Serial No.	Makers	Next Cal. Date	Used
Test Receiver	ESHS10	843276/003	R&S	08.05.22	☒
LISN	ESH3-Z5	100267	R&S	08.07.02	☒
LISN	L2-16A	0000J10705	PMM	-	☒

6.1.3 Measurement uncertainty

Conducted emission measurement : (k=2, 95%)
9kHz-150 kHz : ± 3.263 [dB]
150kHz-30 MHz : ± 2.960 [dB]

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6.1.4 Test data

- BPFR

Frequency	Correction Factor		Line	Quasi-peak			Average		
				Limit	Reading	Result	Limit	Reading	Result
[MHz]	LISN	Cable		[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]
0.201	0.23	0.2	H	79.00	36.51	36.94	66.00	30.64	31.07
0.204	0.10	0.2	N	79.00	34.27	34.57	66.00	29.39	29.69
0.339	0.22	0.2	H	79.00	33.26	33.68	66.00	30.56	30.98
0.402	0.22	0.3	H	79.00	36.74	37.26	66.00	34.01	34.53
0.405	0.10	0.3	N	79.00	36.46	36.86	66.00	34.23	34.63
0.471	0.22	0.3	H	79.00	35.55	36.07	66.00	33.23	33.75
5.770	0.26	0.4	N	73.00	46.71	47.37	60.00	36.42	37.08
5.970	0.39	0.4	H	73.00	52.59	53.38	60.00	40.95	41.74
6.180	0.26	0.4	N	73.00	54.31	54.97	60.00	42.10	42.76
6.383	0.39	0.4	H	73.00	52.21	53.00	60.00	38.04	38.83
12.560	0.67	0.5	H	73.00	49.87	51.04	60.00	36.34	37.51
12.770	0.48	0.5	N	73.00	48.50	49.48	60.00	39.22	40.20

• Note. QP = Quasi-Peak, AV= Average

• Loss = LISN Loss + Cable Loss

• Measurement time : 1 s

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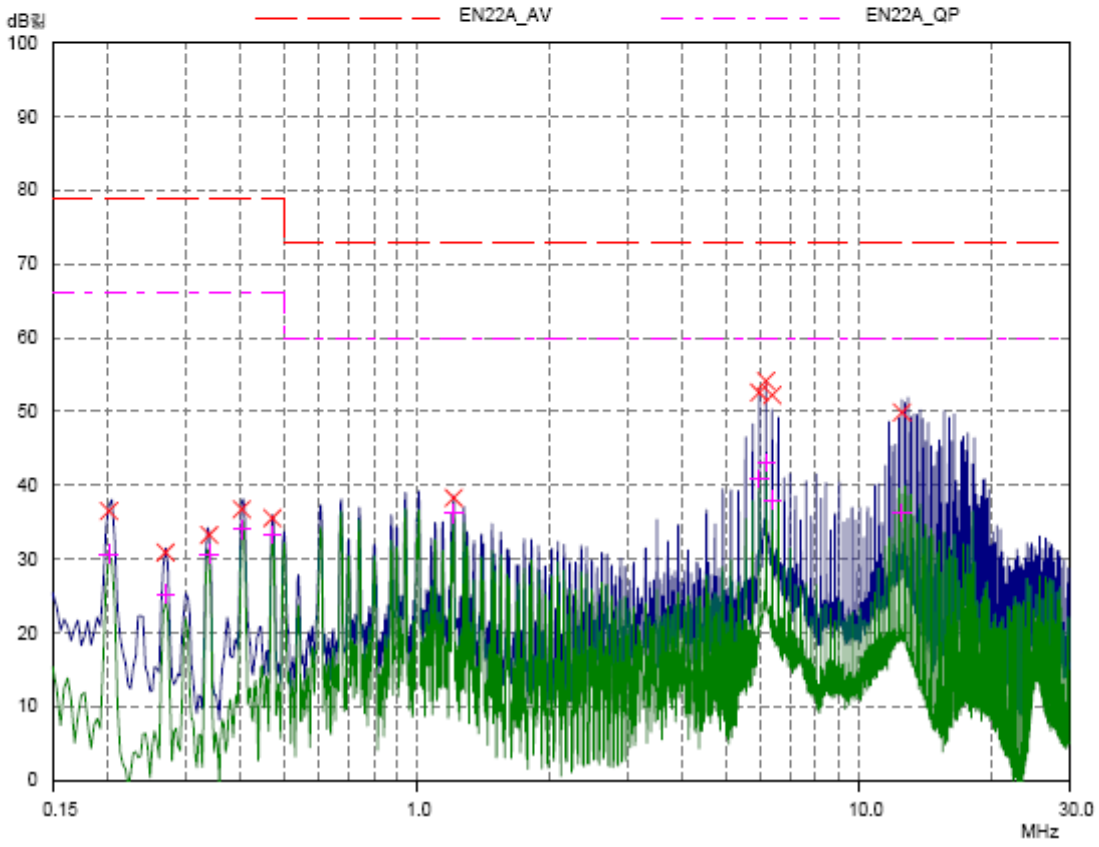
EUT: CONTROLL CARD SET
Manuf: AZIN
Op Cond: H
Operator:
Test Spec: KN22 Class A Conducted Emission
Comment: BIG CARD

Result File: azin-b_h.dat : AZIN BIG CARD_H

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB

Final Measurement: Detectors: X QP / + AV
Meas Time: 1sec
Peaks: 8
Acc Margin: 25 dB



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EUT:	CONTROLL CARD SET
Manuf:	AZIN
Op Cond:	N
Operator:	
Test Spec:	KN22 Class A Conducted Emission
Comment:	BIG CARD

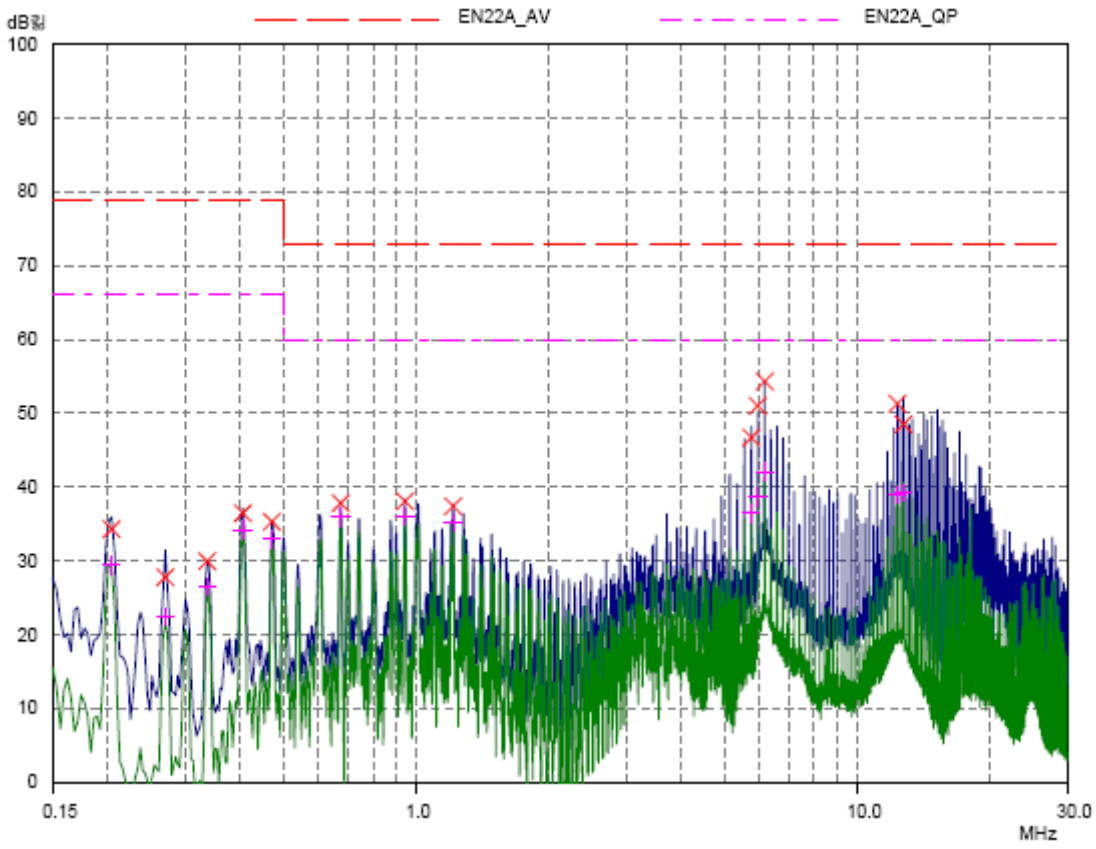
Result File: azin-b_n.dat : AZIN BIG CARD_N

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB

Final Measurement:

Detectors:	X QP / + AV
Meas Time:	1sec
Peaks:	8
Acc Margin:	25 dB



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- BPHR

Frequency	Correction Factor		Line	Quasi-peak			Average		
				Limit	Reading	Result	Limit	Reading	Result
[MHz]	LISN	Cable		[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]
0.201	0.10	0.2	N	79.00	36.39	36.69	66.00	31.23	31.53
0.270	0.23	0.2	H	79.00	31.17	31.60	66.00	25.29	25.72
0.336	0.22	0.2	H	79.00	34.02	34.44	66.00	31.17	31.59
0.405	0.22	0.3	H	79.00	37.33	37.85	66.00	35.12	35.64
0.471	0.10	0.3	N	79.00	35.39	35.79	66.00	32.72	33.12
0.942	0.25	0.3	H	73.00	38.38	38.93	60.00	36.33	36.88
1.011	0.25	0.3	H	73.00	38.95	39.50	60.00	36.69	37.24
5.770	0.26	0.4	N	73.00	47.68	48.34	60.00	36.42	37.08
6.180	0.39	0.4	H	73.00	53.73	54.52	60.00	40.21	41.00
6.390	0.39	0.4	H	73.00	48.10	48.89	60.00	37.12	37.91
12.570	0.48	0.5	N	73.00	48.63	49.61	60.00	37.34	38.32

• Note. QP = Quasi-Peak, AV= Average

• Loss = LISN Loss + Cable Loss

• Measurement time : 1 s

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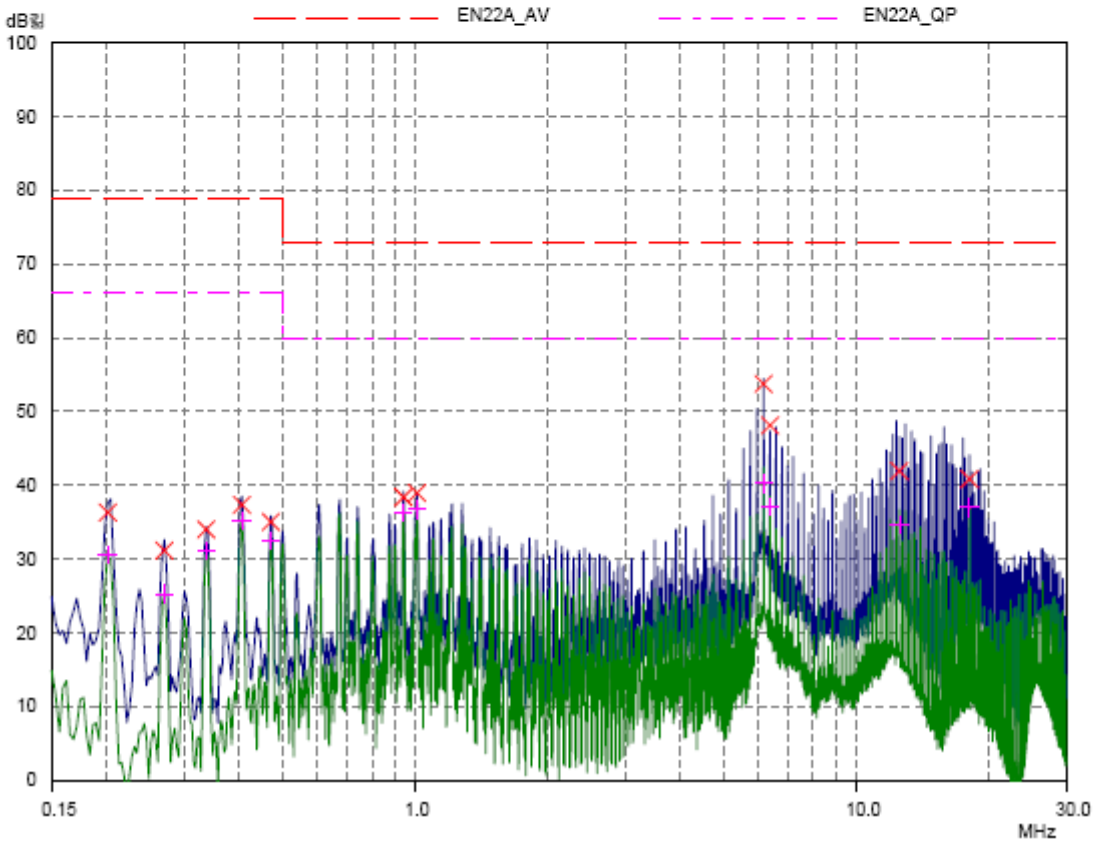
EUT:	CONTROLL CARD SET
Manuf:	AZIN
Op Cond:	H
Operator:	
Test Spec:	KN22 Class A Conducted Emission
Comment:	SMALL CARD
Result File:	azin-s_h.dat : AZIN SMALL CARD_H

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB

Final Measurement:

Detectors:	X QP / + AV
Meas Time:	1sec
Peaks:	8
Acc Margin:	25 dB



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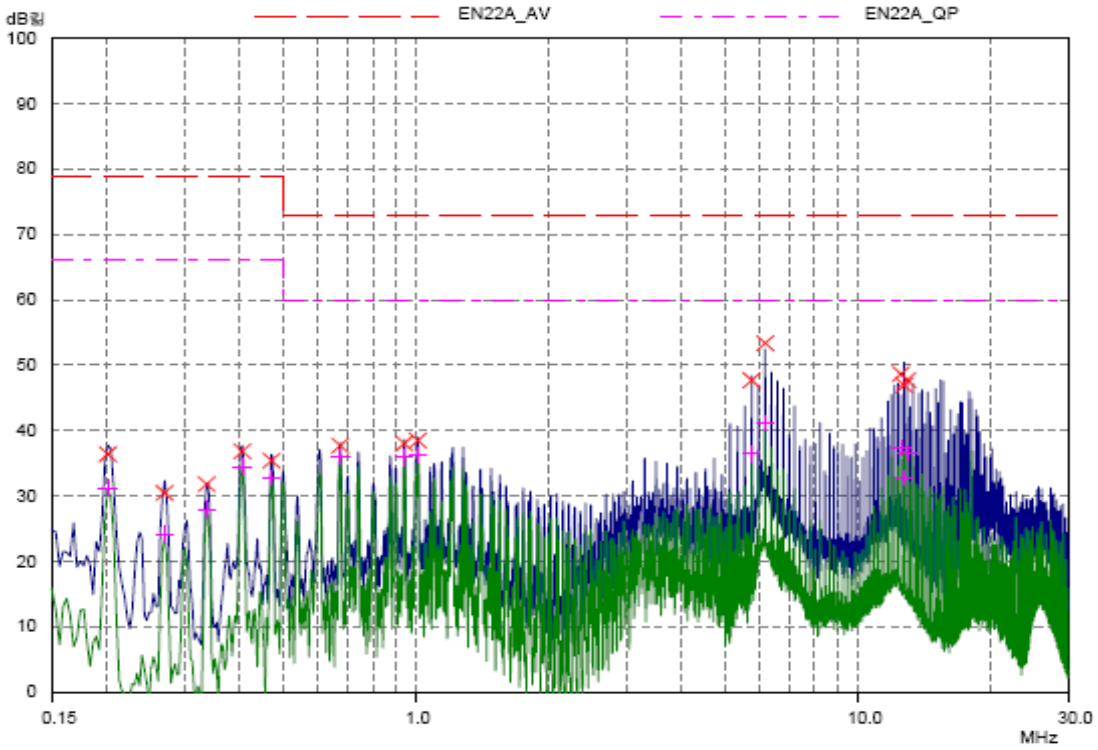
EUT:	CONTROLL CARD SET
Manuf:	AZIN
Op Cond:	N
Operator:	
Test Spec:	KN22 Class A Conducted Emission
Comment:	SMALL CARD
Result File:	azin-s_n.dat : AZIN SMALL CARD_N

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB

Final Measurement:

Detectors:	X QP / + AV
Meas Time:	1sec
Peaks:	8
Acc Margin:	25 dB



6.1.5 Result

Minimum limit margin is 17.25 dB at 6.180 MHz. (BPFR)

Minimum limit margin is 18.48 dB at 6.180 MHz. (BPHR)

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6.2 Radiated emission

6.2.1 Measurement procedure

Mains

A pretest was performed at 3m distance in a semi-anechoic chamber for searching correct frequency. The final test was done at a 10m open area test site with a quasi-peak detector. EUT was placed on a non-metallic table height of 0.8m above the reference ground plane. Cables were folded back and forth forming a bundle 0.3m to 0.4m long and were hanged at a 0.4m height to the ground plane.
Cables connected to EUT were fixed to cause maximum emission. Test was made with the antenna positioned in both the horizontal and vertical planes of polarization.
The measurement antenna was varied in height above the conducting ground plane to obtain the maximum signal strength.

6.2.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next cal. date	Used
Test Receiver	ESVD	841729/010	R&S	08.08.07	☒
TRILOG broadband antenna	VULB 9160	3228	Schwarzbeck Mess-Electronik	08.02.12	☒
Antenna Mast	A109	N/A	DEAIL	-	☒
Turn Table	TS14	N/A	DEAIL	-	☒
10m OATS	-	-	EMC Compliance	-	☒

6.2.3 Measurement uncertainty

Radiated Emission measurement : (k=2, 95%)
30-300 MHz ; 3 m: ±3.721 [dB], 10 m: +3.706, -3.707 [dB]
300-1000 MHz ; 3 m: ±3.818 [dB], 10 m: ±3.802 [dB]

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6.2.4 Test data

- BPFR

Frequency	Reading	Pol.	Height	angle	Correction Factor		Limits	Result	Margin
[MHz]	[dBuV/m]		[m]		Antenna	Cable	[dBuV/m]	[dBuV/m]	[dB]
59.66	12.9	V	1.0	179	11.61	1.23	40.0	25.74	14.26
67.90	16.4	V	1.0	360	10.44	1.28	40.0	28.12	11.88
166.198	17.1	H	3.5	248	12.40	2.05	40.0	31.55	8.45
166.67	19.2	V	2.0	288	12.40	2.05	40.0	33.65	6.35
200.126	21.0	V	1.7	159	9.53	2.32	40.0	32.85	7.15
265.22	17.4	H	4.0	8	11.96	2.69	47.0	32.05	14.95
365.263	16.5	H	2.7	12	14.66	3.20	47.0	34.37	12.63

* Receiving Antenna Mode : Horizontal, Vertical

* 10 m OATS

* Note : Reading = Test Receiver meter,

P= Polarization → H = Horizontal, V = Vertical

Result = Field Strength (Antenna factor + Cable factor + Reading)

- BPHR

Frequency	Reading	Pol.	Height	angle	Correction Factor		Limits	Result	Margin
[MHz]	[dBuV/m]		[m]		Antenna	Cable	[dBuV/m]	[dBuV/m]	[dB]
135.126	19.7	H	2.6	195	11.97	1.87	40.0	33.53	6.47
165.90	20.4	H	3.2	312	12.47	2.05	40.0	34.91	5.09
166.67	18.2	V	2.2	304	12.40	2.05	40.0	32.65	7.35
200.262	20.1	H	3.2	107	9.53	2.32	40.0	31.95	8.05
231.25	22.5	V	2.0	269	10.75	2.51	47.0	35.77	11.24
231.35	21.3	H	2.9	173	10.75	2.51	47.0	34.57	12.44

* Receiving Antenna Mode : Horizontal, Vertical

* 10 m OATS

* Note : Reading = Test Receiver meter,

P= Polarization → H = Horizontal, V = Vertical

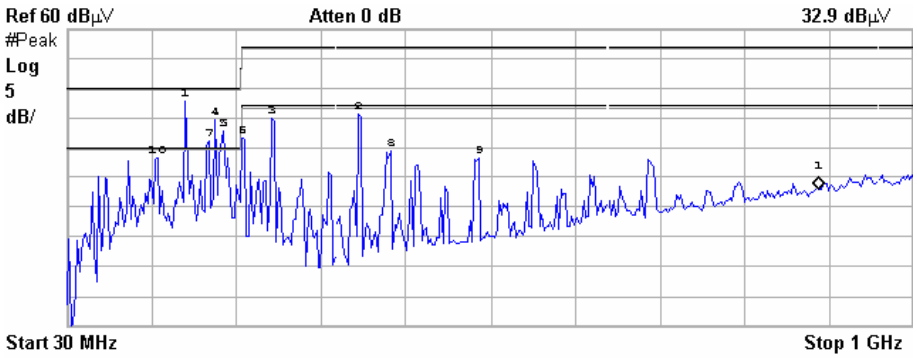
Result = Field Strength (Antenna factor + Cable factor + Reading)

This test report shall not be reproduced except in full, Without the written approval.

***3m Chamber Prescan Data**

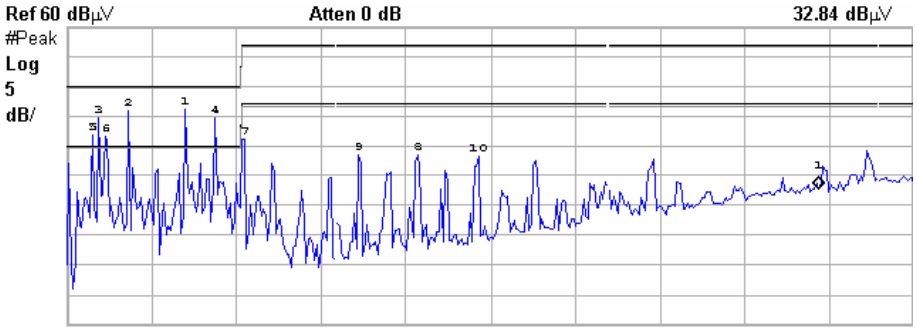
- REC MODE(BPFR)

-Horizontal



Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	166 MHz	47.89 dBµV	6	231 MHz	41.53 dBµV
2	365 MHz	45.6 dBµV	7	192 MHz	41 dBµV
3	265 MHz	45.03 dBµV	8	401 MHz	39.37 dBµV
4	200 MHz	44.8 dBµV	9	500 MHz	38.3 dBµV
5	209 MHz	42.75 dBµV	10	134 MHz	38.21 dBµV

-Vertical



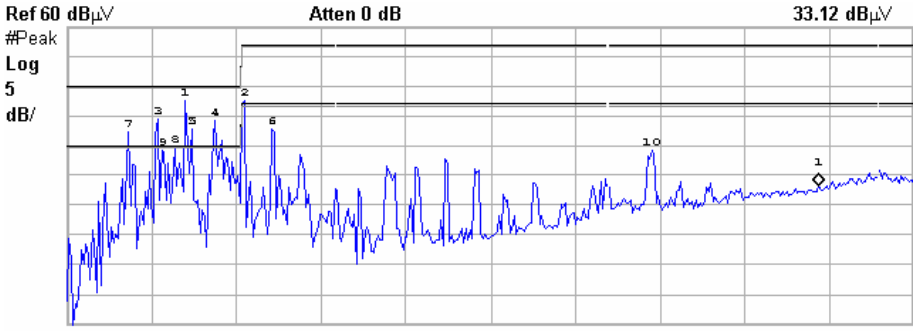
Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	166 MHz	46.11 dBµV	6	74 MHz	41.55 dBµV
2	100 MHz	45.88 dBµV	7	234 MHz	41.14 dBµV
3	66 MHz	44.73 dBµV	8	433 MHz	38.48 dBµV
4	200 MHz	44.61 dBµV	9	365 MHz	38.41 dBµV
5	59 MHz	41.86 dBµV	10	500 MHz	38.23 dBµV

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***3m Chamber Prescan Data**

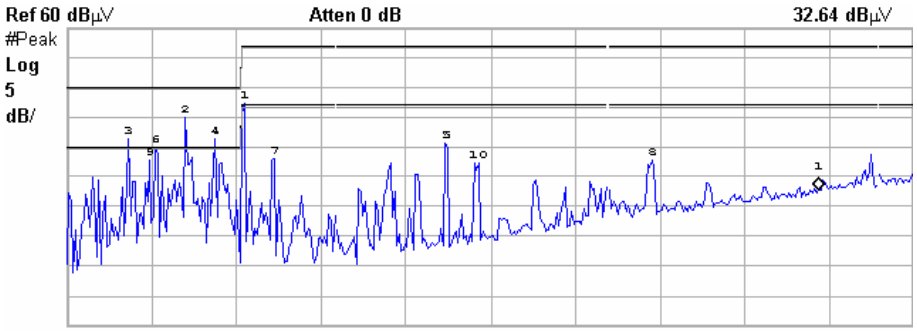
- REC MODE(BPHR)

-Horizontal



Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	166 MHz	47.65 dBµV	6	265 MHz	42.83 dBµV
2	234 MHz	47.63 dBµV	7	100 MHz	42.28 dBµV
3	134 MHz	44.43 dBµV	8	154 MHz	39.73 dBµV
4	200 MHz	44.3 dBµV	9	139 MHz	39.22 dBµV
5	173 MHz	42.86 dBµV	10	699 MHz	39.17 dBµV

-Vertical



Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	234 MHz	47.42 dBµV	6	132 MHz	39.83 dBµV
2	166 MHz	44.82 dBµV	7	268 MHz	38 dBµV
3	100 MHz	41.44 dBµV	8	699 MHz	37.7 dBµV
4	200 MHz	41.32 dBµV	9	125 MHz	37.69 dBµV
5	464 MHz	40.57 dBµV	10	500 MHz	37.3 dBµV

6.2.5. Result

Minimum limit margin is 6.35 dB at 166.67 MHz. (BPFR)

Minimum limit margin is 5.09 dB at 165.90 MHz. (BPHR)

This test report shall not be reproduced except in full, Without the written approval.

6.3 Electrostatic Discharge

6.3.1 Measurement procedure

A ground reference plane was located on the floor, and connected to earth via a low impedance connection. The return cable of the ESD generator was connected to the reference plane. In case of floor standing equipment, EUT was placed on the reference plane on 0.1 m of insulating Support. In case of table top equipment, EUT was placed on a wooden table 0.8m above the reference grounded floor. A horizontal coupling plane(HCP) was placed on the table, and Connected to the reference plane via a 470㏀ resistor located in each end (0.5mm insulating support between EUT and HCP). In both cases a vertical coupling plane(VCP) OF 0.5 X 0.5m was located 10cm from the EUT's sides. The VCP was connected to the reference plane in the same matter as the HCP.

6.3.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. Date	Used
ESD Tester	ESS-100L	5099C02066	NOISE KEN	08.06.19	☒
HCP	-	-	-	-	☒
VCP	-	-	-	-	☒

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6.3.3 Test Data

Test Specification : EN61000-4-2

Kind of Discharges

☒ Contact Discharge (Direct Discharge)

☐ Air Discharge

☒ HCP / VCP (Indirect Discharge)

Discharge Voltages

☒ Contact Discharge : ± 2, 4kV

☐ Air Discharge :

Discharge Impedance

☒ 330 Ω/150pF ☐ 2KΩ/330pF

Number Of Discharge

☒ Number of discharges per point, for each voltage and polarity
: 50 (Interval between discharges : ≥ 1s)

Test point (Please refer to attached photograph.)

▪Contact Discharge : Screw, Bleakest, Port

▪Air Discharge : No air discharge part

Test Results

☒ Complied ☐ Not complied

Comment :

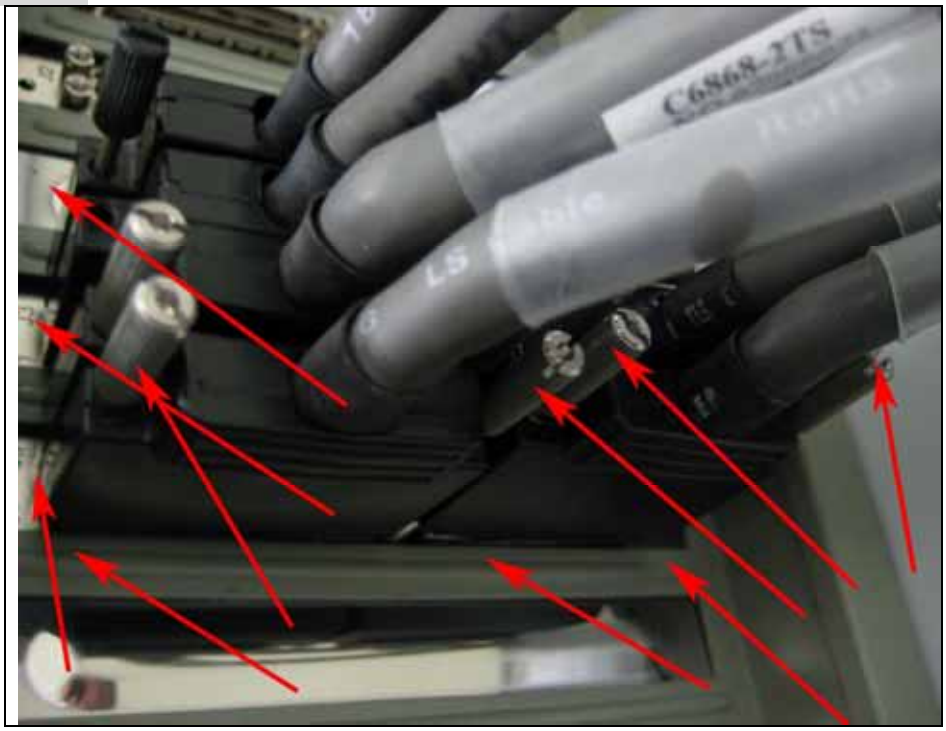
-There was no change of operation status during above testing.

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Electrostatic Discharge (Test Point)

Air discharge →
Contact discharge →

- **BPHR**



- **BPHR**



This test report shall not be reproduced except in full, Without the written approval.

6.4 Radio Frequency Electromagnetic Fields

6.4.1 Measurement procedure

The test was performed at 3m full anechoic chamber.
For floor standing equipment, the EUT was standing on the floor.
For tabletop equipment, the EUT was located on a wooden table 0.8m above the floor.
The EUT was tested all sides, horizontal and vertical polarization.
The field uniformity was calibrated for 1V/m, 3V/m, 10V/m.

6.4.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Power meter	PM2002	302852	AR	08.05.03	☒
Field monitor	SI-300	-	TDK	-	☒
Power sensor (with adapter)	PH2000	303224	AR	08.05.03	☒
Power sensor (with adapter)	PH2000	303222	AR	08.05.03	☒
Directional coupler	DC6180	303976	AR	08.05.03	☒
Amplifier	150W1000M2	303843	AR	08.05.03	☒
Amplifier	60S1G3M2	320444	AR	08.04.21	☒
Signal generator	E4421B	GB40052295	AGILENT	08.10.25	☒
BiconiLog Ant.	LPDA-0803	130269	ETS	08.03.17	☒
Directional Coupler	DC7144M1	320279	AR	08.01.18	☒

6.4.3 Measurement uncertainty

Radio Frequency Electromagnetic Fields : ± 1.0556 [dB](k=2, 95%)

This test report shall not be reproduced except in full, Without the written approval.

6.4.4 Test Data

Test Specification : EN 61000-4-3

Frequency Range

☒ 80MHz – 1000MHz ☐ 900 MHz ± 5MHz ☐ 26MHz - 500MHz

Test level

☐1V/m ☒ 3V/m ☐ 10V/m

Modulation

☒ AM : 1kHz, 80%

☐ PM :

Frequency step

☒ log 1% step ☐ log 3% step ☐ log 5% step

Dwell Time

☒ 3 s ☐ 2 s ☐ 1 s

Test point

☒ Front

☒ Rear

☒ Left

☒ Right

Test Results

☒ Complied ☐ Not complied

Comment :

- There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.5 Electric Fast Transient/BURST

6.5.1 Measurement procedure

A ground reference plane was located on the floor.
EFT generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.8 m wooden table.
For tabletop equipment, EUT was placed on a wooden table(0.8m) above the reference plane.

6.5.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
EFT/B Tester	UCS 500 M6	V0545100858	EM TEST	08.01.03	☒
Capacitive coupling clamp	N/A	N/A	EM TEST	-	☐

This test report shall not be reproduced except in full, Without the written approval.

6.5.3 Test Data

Test Specification : EN 61000-4-4

Coupling

☒ Power ☐ Signal Lines ☐ Telecommunication line

Test level

☒ Power : ±1 kV

☐ Signal Line :

☐ Tel. line :

Test mode

- Power : L1, L2, PE, L1+L2, L1+PE, L2+PE, L1+L2+PE

-Signal Line :

Burst frequency : 5 kHz, 5/50 ns

Coupling Time : > 60 s

Test Results

☒ Complied ☐ Not complied

Comment :

- There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.6 Surge

6.6.1 Measurement procedure

A ground reference plane was located on the floor.
SURGE generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.8 m wooden table.
For tabletop equipment, EUT was placed on a wooden table(0.8m) above the reference plane.

6.6.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Surge Generator	UCS 500 M6	V0545100858	EM TEST	08.01.03	☒
CDN	CNV 508 S2	0402-01	EM TEST	-	☐

This test report shall not be reproduced except in full, Without the written approval.

6.6.3 Test Data

Test Specification : EN 61000-4-5

Coupling

☒ Power ☐ Signal Line ☐ Telecommunication line

Test level

☒ Power : L-N : ± 1 kV
L-PE, N-PE : ± 2 kV

☐ Signal Line :

☐ Tel. line :

Test mode

- AC Power : L-N, L-PE, N-PE

-Signal Line :

Coupling Impedance

☐ 40Ω+0.5μF ☐ 40Ω ☒ 10Ω+9μF ☒ 18μF

Coupling Time : ＞ 1 min

Number of Surge : 5

Angle : 0, 90, 180, 270

Test Results

☒ Complied ☐ Not complied

Comment :

-There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.7 Conducted Immunity

6.7.1 Measurement procedure

A ground reference plane was located on the floor.
For tabletop equipment, the test was performed on a ground reference plane on a 0.8m wooden table.
The EUT was isolated 0.1 m isolating support.
The ground plane was connected to floor reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.8 m wooden table.
This test were Performed using CDN for mains, clamp for signal and injection probe.

6.7.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
CS generator	NSG 2070	1054	EM TEST	08.05.03	☒
CDN	M016	16674	EM TEST	08.04.10	☒
Attenuator	INA2070-1	2054	Schaffner	-	☒
EM Clamp	KEMZ 801	17643	Schaffner	-	☐
EM Clamp	KEMA 801	17899	Scahffner	-	☐
Current probe	MD720	W1345167/M6/0068	Schaffner	-	☐

This test report shall not be reproduced except in full, Without the written approval.

6.7.3 Test Data

Test Specification : EN 61000-4-6

Frequency Range

☒ 150 kHz – 80MHz ☐ 150 kHz - 230MHz ☐150 kHz - 500MHz

Test point: Power

Coupling

☒ Power : CDN

☐ Signal :

☐ Tel. line :

Test level

☐ 1V ☒ 3V ☐ 10V

Modulation

☒ AM : 1kHz, 80%

☐ PM : 1Hz (0.5 s ON : 0.5 s OFF)

Frequency step

☒ log 1% step ☐ log 3% step ☐ log 5% step

Dwell Time

☒ 3 s ☐ 2 s ☐ 1 s

Test Results

☒ Complied ☐ Not complied

Comment :

-There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.8 Dips and Interruptions

6.8.1 Measurement procedure

The dips/interruption test is only applicable to AC mains.
The dips/interruptions were applied at zero crossing.

6.8.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
dips/interruption Tester	UCS 500 M6	V0545100858	EM TEST	08.01.03	☒

6.8.3 Test data

Test specification : EN 61000-4-11

Test data

- 230Vac

Test Level (%UT)	Dip/Int. (%UT)	Duration /Period	Phase (°)	Count number	Result
0%	100%	0.5 Period	0 / 180	3T	A
70%	30%	25 Period	0 / 180	3T	A
0 %	100 %	250 Period	0 / 180	3T	C

Test results

☒ Complied ☐ Not complied

Comment :

- There was no change of operation status during above testing.
- The power of EUT is off during the interruption test. After the test, EUT is getting back to normal operation. (250 Period)

This test report shall not be reproduced except in full, Without the written approval.

6.9 Harmonics

6.9.1 Measurement procedure

The equipment is supplied in series with shunt(s) Rm or current transformer(s) from a source having the same nominal voltage and frequency as the rated supply voltage and frequency of the equipment.

Measurements shall be made under normal load, or conditions for adequate heat discharge, and under normal operating conditions.

User's operation controls or automatic programmers shall be set to produce the maximum harmonic component, for each successive harmonic component in turn. For the purpose of harmonic current limitation, equipment is classified as follows :

Class A : Equipment not specified in one of the three other Classes shall be considered as Class A equipment.

- Balanced three-phase equipment;
- Household appliances excluding equipment identified as Class D;
- Tools excluding portable tools;
- Dimmers for incandescent lamps;
- Audio equipment.

Class B : Portable tools; Arc welding equipment which is not professional equipment.

Class C : Lighting equipment.

Class D : Equipment having a specified power according to 6.2.2 less than or equal to 600 w,
of the following types:
- Personal computers and personal computer monitors;
- Television receivers.

6.9.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Harmonics/Flicker meter	5001x-CTS-400-413	54984	C.I.	08.05.03	☒
Test site	Immunity area	-	-	-	☒

6.9.3 Test data

- Refer to attached test data

Test results

☒ Complied ☐ Not complied

This test report shall not be reproduced except in full, Without the written approval.

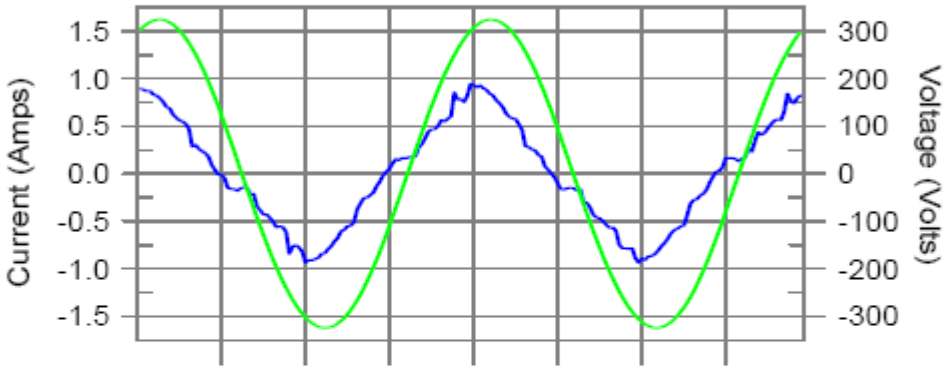
- BPFR

Harmonics – Class-A per Ed. 3.0 (2005-11)(Run time) incl. inter-harmonics

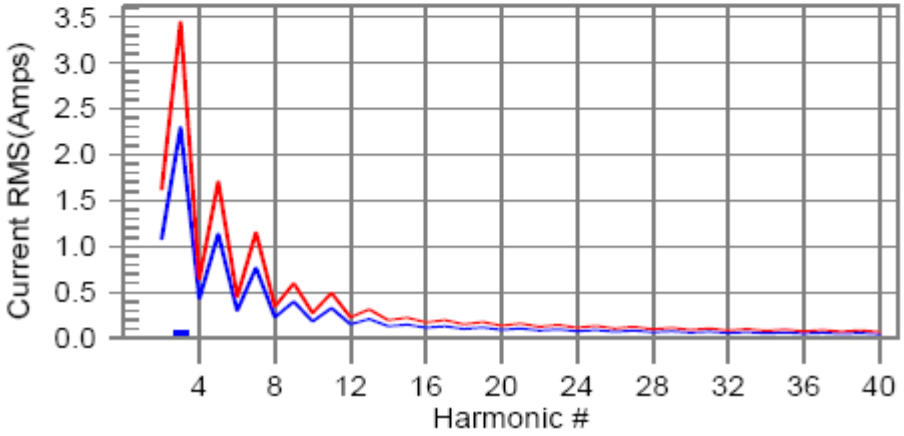
EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-11-02 Start time: 오후 5:50:44 End time: 오후 5:53:25
Test duration (min): 2.5 Data file name: H-000373.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Current & voltage waveforms



Harmonics and Class A limit line European Limits



Test result: Pass Worst harmonic was #17 with 8.45% of the limit.

This test report shall not be reproduced except in full, Without the written approval.

Current Test Result Summary (Run time)

EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-11-02 Start time: 오후 5:50:44 End time: 오후 5:53:25
Test duration (min): 2.5 Data file name: H-000373.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal
THC(A): 0.09 I-THD(%): 16.75 POHC(A): 0.019 POHC Limit(A): 0.251
Highest parameter values during test:

V_RMS (Volts):	229.65	Frequency(Hz):	50.00
I_Peak (Amps):	0.986	I_RMS (Amps):	0.546
I_Fund (Amps):	0.526	Crest Factor:	1.836
Power (Watts):	117.9	Power Factor:	0.941

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.002	1.080	0.2	0.003	1.620	0.16	Pass
3	0.079	2.300	3.4	0.080	3.450	2.33	Pass
4	0.001	0.430	0.2	0.001	0.645	0.19	Pass
5	0.009	1.140	0.8	0.009	1.710	0.55	Pass
6	0.000	0.300	0.2	0.001	0.450	0.14	Pass
7	0.012	0.770	1.5	0.012	1.155	1.03	Pass
8	0.000	0.230	0.2	0.000	0.345	0.14	Pass
9	0.011	0.400	2.6	0.011	0.600	1.85	Pass
10	0.000	0.184	0.3	0.001	0.276	0.26	Pass
11	0.014	0.330	4.1	0.014	0.495	2.85	Pass
12	0.001	0.153	0.3	0.001	0.230	0.29	Pass
13	0.004	0.210	2.1	0.005	0.315	1.44	Pass
14	0.001	0.131	0.4	0.001	0.197	0.43	Pass
15	0.005	0.150	3.5	0.006	0.225	2.58	Pass
16	0.001	0.115	0.7	0.001	0.173	0.62	Pass
17	0.017	0.132	12.5	0.017	0.199	8.45	Pass
18	0.001	0.102	0.8	0.001	0.153	0.62	Pass
19	0.006	0.118	4.7	0.006	0.178	3.33	Pass
20	0.001	0.092	0.8	0.001	0.138	0.64	Pass
21	0.013	0.107	11.9	0.013	0.161	8.10	Pass
22	0.001	0.084	0.8	0.001	0.125	0.73	Pass
23	0.008	0.098	8.2	0.008	0.147	5.65	Pass
24	0.001	0.077	0.9	0.001	0.115	0.70	Pass
25	0.004	0.090	3.9	0.004	0.135	2.75	Pass
26	0.001	0.071	1.3	0.001	0.106	0.95	Pass
27	0.005	0.083	5.7	0.005	0.125	4.16	Pass
28	0.001	0.066	1.4	0.001	0.099	1.12	Pass
29	0.004	0.078	5.3	0.004	0.116	3.72	Pass
30	0.001	0.061	1.3	0.001	0.092	1.07	Pass
31	0.001	0.073	1.4	0.001	0.109	1.12	Pass
32	0.001	0.058	1.4	0.001	0.086	1.11	Pass
33	0.004	0.068	6.5	0.005	0.102	4.67	Pass
34	0.001	0.054	1.5	0.001	0.081	1.14	Pass
35	0.005	0.064	8.0	0.005	0.096	5.57	Pass
36	0.001	0.051	1.5	0.001	0.077	1.21	Pass
37	0.006	0.061	9.4	0.006	0.091	6.46	Pass
38	0.001	0.048	1.3	0.001	0.073	1.19	Pass
39	0.001	0.058	1.9	0.001	0.087	1.66	Pass
40	0.001	0.046	1.3	0.001	0.069	1.05	Pass

This test report shall not be reproduced except in full, Without the written approval.

Voltage Source Verification Data (Run time)

EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-11-02 Start time: 오후 5:50:44 End time: 오후 5:53:25
Test duration (min): 2.5 Data file name: H-000373.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Highest parameter values during test:

Voltage (Vrms):	229.65	Frequency(Hz):	50.00
I_Peak (Amps):	0.986	I_RMS (Amps):	0.546
I_Fund (Amps):	0.526	Crest Factor:	1.836
Power (Watts):	117.9	Power Factor:	0.941

Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.083	0.459	18.12	OK
3	0.647	2.067	31.30	OK
4	0.081	0.459	17.56	OK
5	0.063	0.919	6.85	OK
6	0.061	0.459	13.34	OK
7	0.060	0.689	8.66	OK
8	0.043	0.459	9.46	OK
9	0.039	0.459	8.45	OK
10	0.032	0.459	7.03	OK
11	0.053	0.230	23.24	OK
12	0.033	0.230	14.21	OK
13	0.031	0.230	13.53	OK
14	0.023	0.230	10.22	OK
15	0.033	0.230	14.15	OK
16	0.034	0.230	14.99	OK
17	0.034	0.230	14.70	OK
18	0.022	0.230	9.78	OK
19	0.016	0.230	6.95	OK
20	0.024	0.230	10.60	OK
21	0.019	0.230	8.30	OK
22	0.014	0.230	6.19	OK
23	0.022	0.230	9.74	OK
24	0.017	0.230	7.25	OK
25	0.016	0.230	6.98	OK
26	0.020	0.230	8.50	OK
27	0.012	0.230	5.28	OK
28	0.012	0.230	5.38	OK
29	0.022	0.230	9.38	OK
30	0.024	0.230	10.50	OK
31	0.013	0.230	5.63	OK
32	0.011	0.230	4.87	OK
33	0.017	0.230	7.26	OK
34	0.021	0.230	9.33	OK
35	0.020	0.230	8.51	OK
36	0.011	0.230	4.78	OK
37	0.018	0.230	7.93	OK
38	0.013	0.230	5.55	OK
39	0.015	0.230	6.40	OK
40	0.011	0.230	4.68	OK

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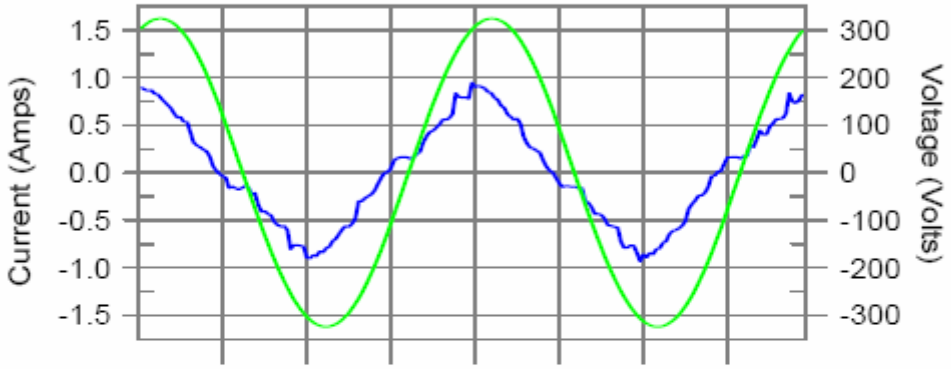
- BPHR

Harmonics – Class-A per Ed. 3.0 (2005-11)(Run time) incl. inter-harmonics

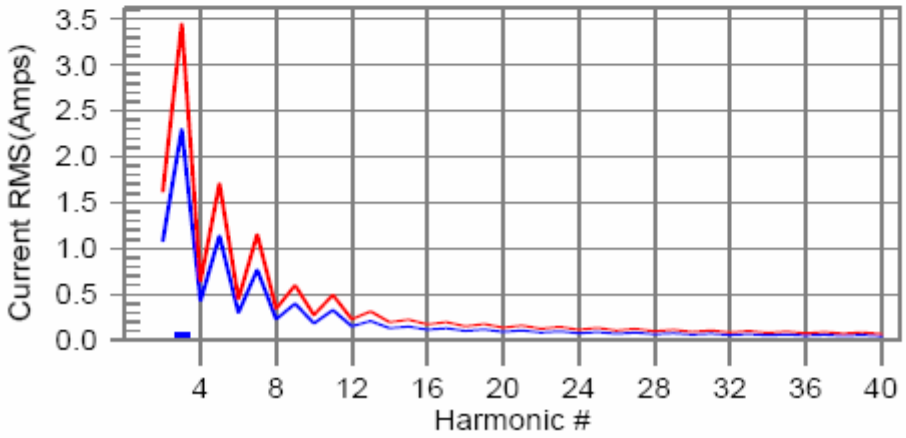
EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-11-02 Start time: 오후 6:17:35 End time: 오후 6:20:16
Test duration (min): 2.5 Data file name: H-000376.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Current & voltage waveforms



Harmonics and Class A limit line European Limits



Test result: Pass Worst harmonic was #17 with 8.47% of the limit.

This test report shall not be reproduced except in full, Without the written approval.

Current Test Result Summary (Run time)

EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-11-02 Start time: 오후 6:17:35 End time: 오후 6:20:16
Test duration (min): 2.5 Data file name: H-000376.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal
THC(A): 0.09 I-THD(%): 16.52 POHC(A): 0.019 POHC Limit(A): 0.251
Highest parameter values during test:

V_RMS (Volts):	229.65	Frequency(Hz):	50.00
I_Peak (Amps):	0.984	I_RMS (Amps):	0.541
I_Fund (Amps):	0.528	Crest Factor:	1.830
Power (Watts):	116.8	Power Factor:	0.940

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.002	1.080	0.2	0.003	1.620	0.16	Pass
3	0.079	2.300	3.4	0.080	3.450	2.31	Pass
4	0.001	0.430	0.2	0.001	0.645	0.19	Pass
5	0.009	1.140	0.8	0.009	1.710	0.54	Pass
6	0.000	0.300	0.2	0.001	0.450	0.14	Pass
7	0.012	0.770	1.5	0.012	1.155	1.05	Pass
8	0.000	0.230	0.2	0.001	0.345	0.16	Pass
9	0.011	0.400	2.7	0.011	0.600	1.83	Pass
10	0.000	0.184	0.2	0.001	0.276	0.24	Pass
11	0.014	0.330	4.1	0.014	0.495	2.79	Pass
12	0.001	0.153	0.3	0.001	0.230	0.28	Pass
13	0.005	0.210	2.2	0.005	0.315	1.54	Pass
14	0.001	0.131	0.4	0.001	0.197	0.37	Pass
15	0.006	0.150	3.7	0.006	0.225	2.56	Pass
16	0.001	0.115	0.7	0.001	0.173	0.54	Pass
17	0.017	0.132	12.5	0.017	0.199	8.47	Pass
18	0.001	0.102	0.8	0.001	0.153	0.59	Pass
19	0.005	0.118	4.5	0.006	0.178	3.18	Pass
20	0.001	0.092	0.8	0.001	0.138	0.62	Pass
21	0.013	0.107	12.1	0.013	0.161	8.22	Pass
22	0.001	0.084	0.8	0.001	0.125	0.70	Pass
23	0.008	0.098	8.4	0.008	0.147	5.70	Pass
24	0.001	0.077	0.9	0.001	0.115	0.71	Pass
25	0.004	0.090	3.9	0.004	0.135	2.75	Pass
26	0.001	0.071	1.3	0.001	0.106	0.97	Pass
27	0.005	0.083	5.9	0.005	0.125	4.27	Pass
28	0.001	0.066	1.4	0.001	0.099	1.10	Pass
29	0.004	0.078	5.1	0.004	0.116	3.61	Pass
30	0.001	0.061	1.3	0.001	0.092	1.07	Pass
31	0.001	0.073	1.4	0.001	0.109	1.06	Pass
32	0.001	0.058	1.4	0.001	0.086	1.14	Pass
33	0.004	0.068	6.4	0.005	0.102	4.58	Pass
34	0.001	0.054	1.4	0.001	0.081	1.11	Pass
35	0.005	0.064	8.0	0.005	0.096	5.55	Pass
36	0.001	0.051	1.5	0.001	0.077	1.17	Pass
37	0.006	0.061	9.4	0.006	0.091	6.41	Pass
38	0.001	0.048	1.3	0.001	0.073	1.14	Pass
39	0.001	0.058	2.2	0.001	0.087	1.72	Pass
40	0.001	0.046	1.3	0.001	0.069	1.01	Pass

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Voltage Source Verification Data (Run time)

EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-11-02 Start time: 오후 6:17:35 End time: 오후 6:20:16
Test duration (min): 2.5 Data file name: H-000376.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Highest parameter values during test:

Voltage (Vrms):	229.65	Frequency(Hz):	50.00
I_Peak (Amps):	0.984	I_RMS (Amps):	0.541
I_Fund (Amps):	0.528	Crest Factor:	1.830
Power (Watts):	116.8	Power Factor:	0.940

Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.084	0.459	18.35	OK
3	0.646	2.067	31.25	OK
4	0.082	0.459	17.81	OK
5	0.064	0.919	6.94	OK
6	0.060	0.459	13.09	OK
7	0.061	0.689	8.81	OK
8	0.042	0.459	9.25	OK
9	0.038	0.459	8.34	OK
10	0.034	0.459	7.37	OK
11	0.055	0.230	23.97	OK
12	0.035	0.230	15.36	OK
13	0.032	0.230	13.94	OK
14	0.024	0.230	10.54	OK
15	0.032	0.230	14.11	OK
16	0.036	0.230	15.81	OK
17	0.034	0.230	14.88	OK
18	0.024	0.230	10.45	OK
19	0.015	0.230	6.65	OK
20	0.024	0.230	10.46	OK
21	0.018	0.230	7.98	OK
22	0.015	0.230	6.60	OK
23	0.022	0.230	9.43	OK
24	0.017	0.230	7.27	OK
25	0.014	0.230	6.23	OK
26	0.019	0.230	8.39	OK
27	0.012	0.230	5.40	OK
28	0.011	0.230	4.98	OK
29	0.022	0.230	9.77	OK
30	0.025	0.230	11.09	OK
31	0.014	0.230	5.93	OK
32	0.012	0.230	5.19	OK
33	0.017	0.230	7.33	OK
34	0.020	0.230	8.90	OK
35	0.020	0.230	8.79	OK
36	0.011	0.230	4.86	OK
37	0.016	0.230	6.97	OK
38	0.011	0.230	4.95	OK
39	0.015	0.230	6.48	OK
40	0.011	0.230	4.90	OK

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6.10 Flicker

6.10.1 Measurement procedure

EUT was connected to the power analyzer system.
Measurement was performed to obtain the desired flicker parameters.
The measuring time depends on which parameters are to be measured.
P_{lt} = 2 h
P_{st} = 10 min
Controls and automatic programs shall be set to produce the most unfavorable sequence of voltage changes, using only those combinations of controls and programs are mentioned by the manufacturer in the instruction manual.

6.10.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Harmonics/Flicker meter	5001x-CTS-400-413	54984	C.I.	08.05.03	☒
Test site	Immunity area	-	-	-	☒

6.10.3 Test data

- Refer to attached test data

Test results

☒ Complied ☐ Not complied

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- BPFR

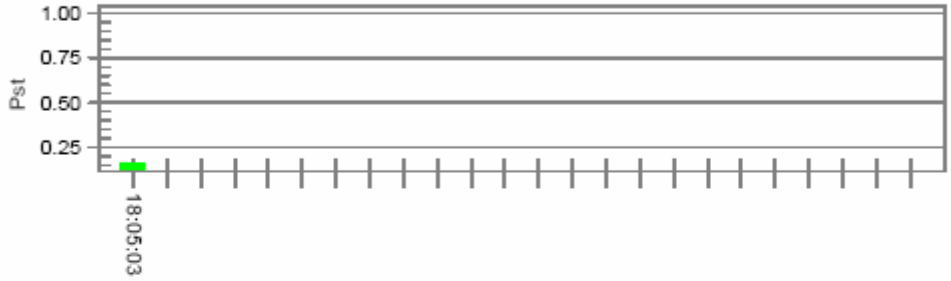
Flicker Test Summary per EN/IEC61000-3-3 (Run time)

EUT: Equipment Under Test
Test category: All parameters (European limits)
Test date: 2007-11-02
Test duration (min): 10
Comment: Comments
Customer: Customer

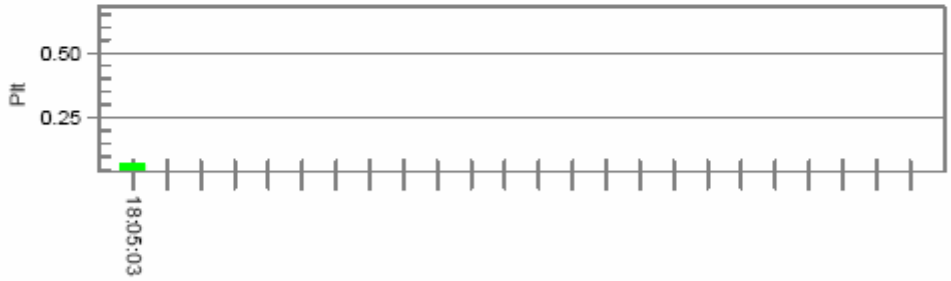
Tested by: Test Operator
Test Margin: 100
Start time: 오후 5:54:43
End time: 오후 6:05:05
Data file name: F-000374.cts_data

Test Result: Pass
Status: Test Completed

Pst and limit line
European Limits



Plt and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt):	229.41			
Highest dt (%):	0.00	Test limit (%):	3.30	Pass
Time(mS) > dt:	0.0	Test limit (mS):	500.0	Pass
Highest dc (%):	0.00	Test limit (%):	3.30	Pass
Highest dmax (%):	0.00	Test limit (%):	4.00	Pass
Highest Pst (10 min. period):	0.160	Test limit:	1.000	Pass
Highest Plt (2 hr. period):	0.070	Test limit:	0.650	Pass

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- BPHR

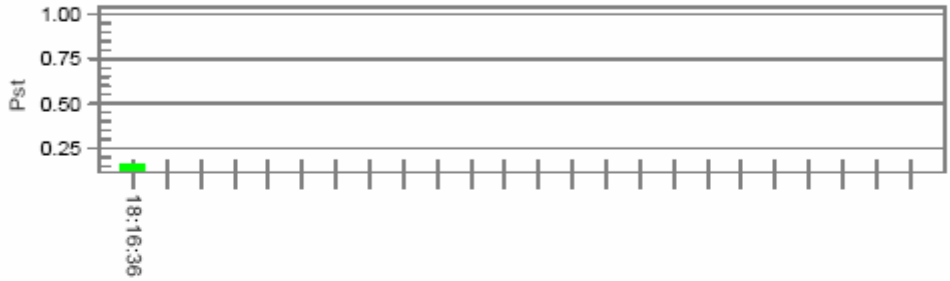
Flicker Test Summary per EN/IEC61000-3-3 (Run time)

EUT: Equipment Under Test
Test category: All parameters (European limits)
Test date: 2007-11-02 Start time: 오후 6:06:16 End time: 오후 6:16:37
Test duration (min): 10 Data file name: F-000375.cts_data
Comment: Comments
Customer: Customer

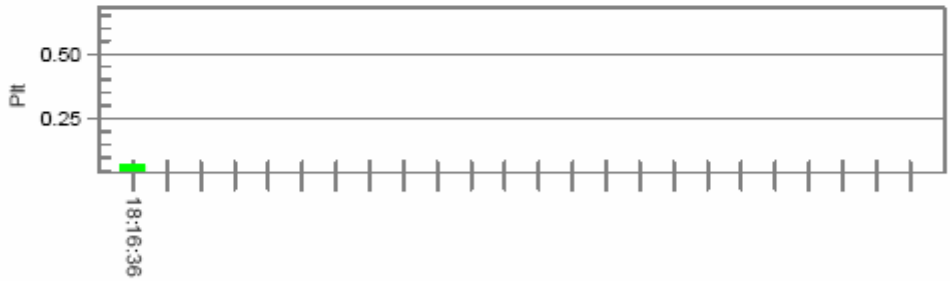
Tested by: Test Operator
Test Margin: 100

Test Result: Pass Status: Test Completed

Pst and limit line European Limits



Plt and limit line



Parameter values recorded during the test:

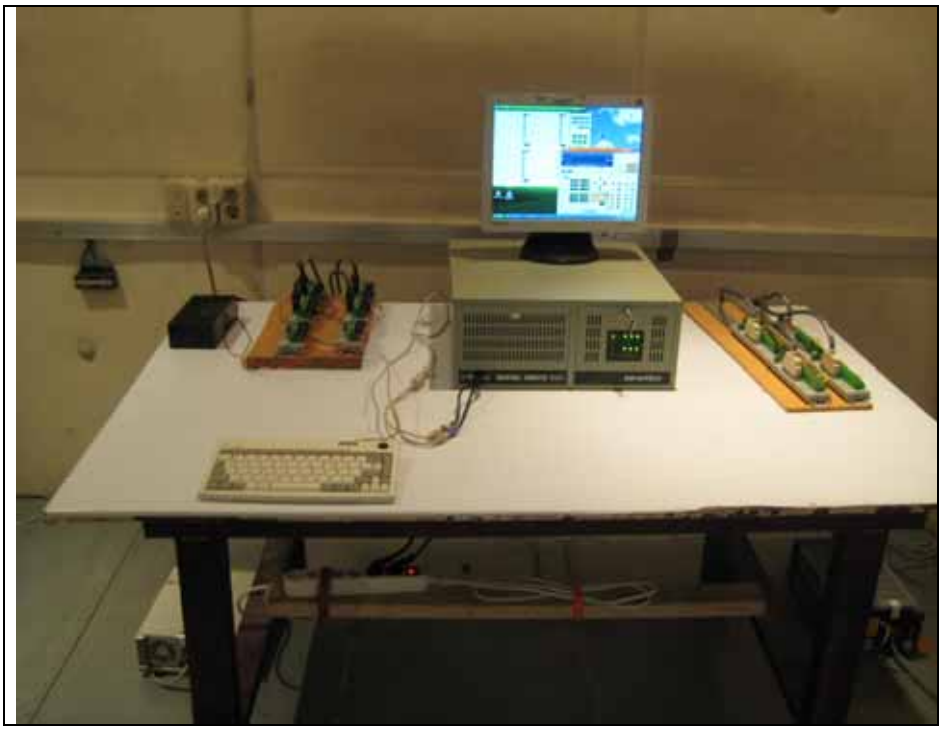
Vrms at the end of test (Volt):	229.38			
Highest dt (%):	0.00	Test limit (%):	3.30	Pass
Time(mS) > dt:	0.0	Test limit (mS):	500.0	Pass
Highest dc (%):	0.00	Test limit (%):	3.30	Pass
Highest dmax (%):	0.00	Test limit (%):	4.00	Pass
Highest Pst (10 min. period):	0.160	Test limit:	1.000	Pass
Highest Plt (2 hr. period):	0.070	Test limit:	0.650	Pass

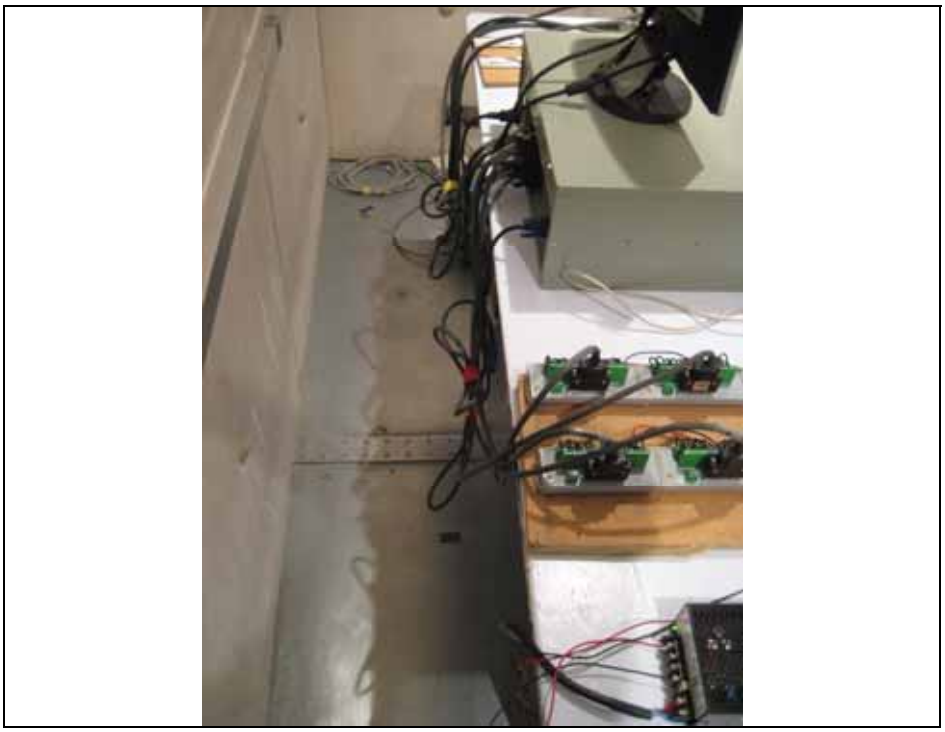
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7. Test photographs

Conducted emission

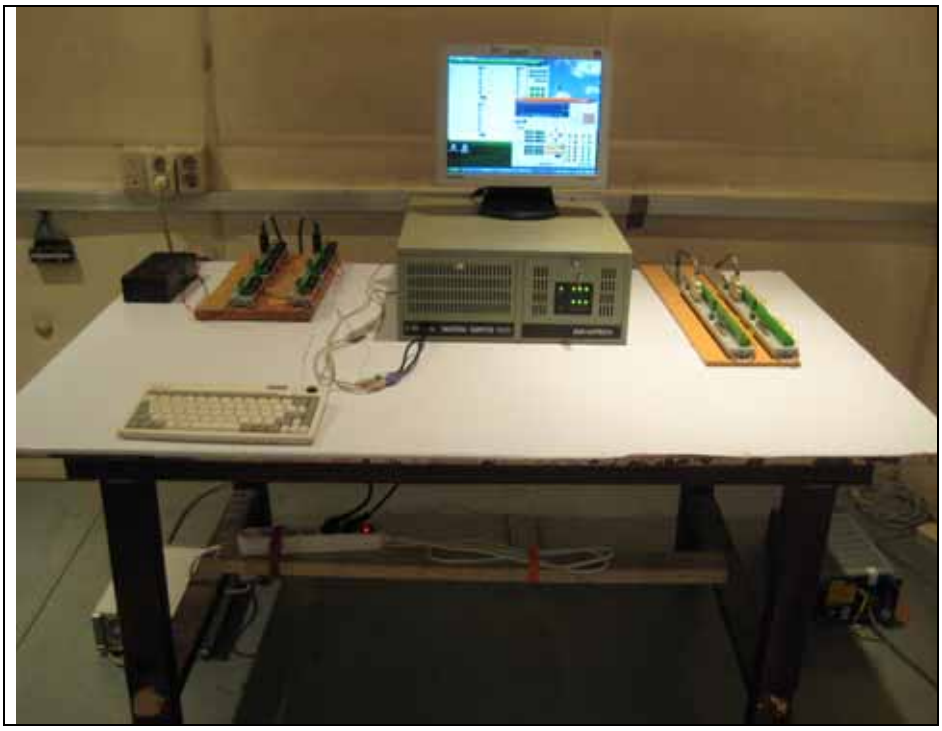
- BPFR





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- BPHR





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Radiated Emission

- BPFR





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- BPHR





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Electrostatic Discharge

- BPFR



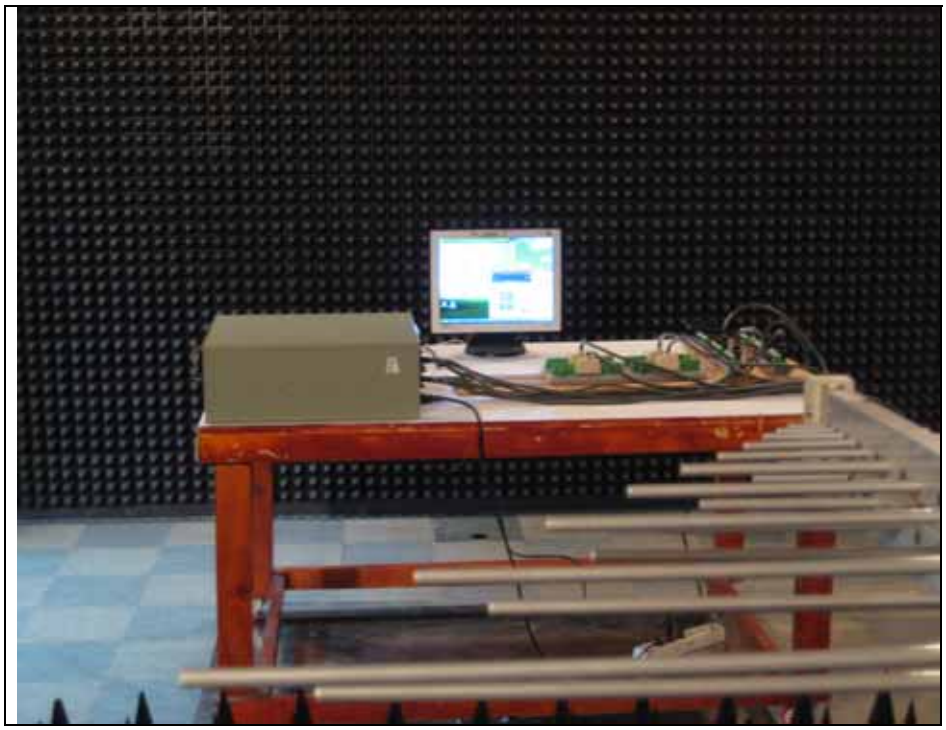
- BPHR



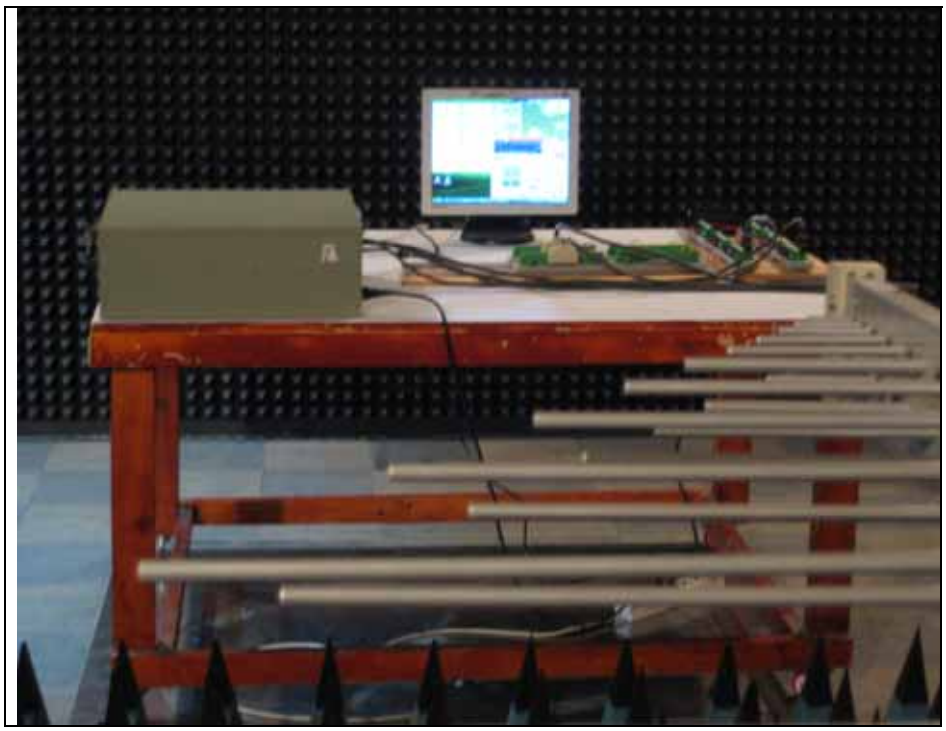
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Radio Frequency Electromagnetic Fields

- BPFR



- BPHR



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Electric Fast Transient

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Surge

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Conducted Immunity

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Voltage dip/interruption

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Harmonics/Flicker

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8. E.U.T. photographs

-BPFR

Front View



Rear View



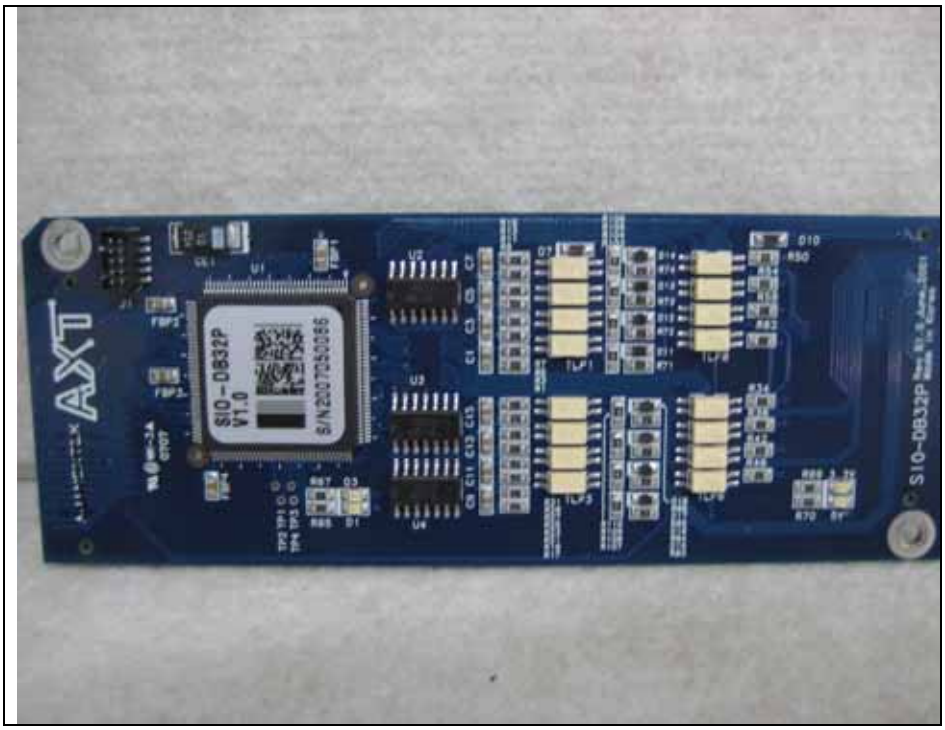
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Inside



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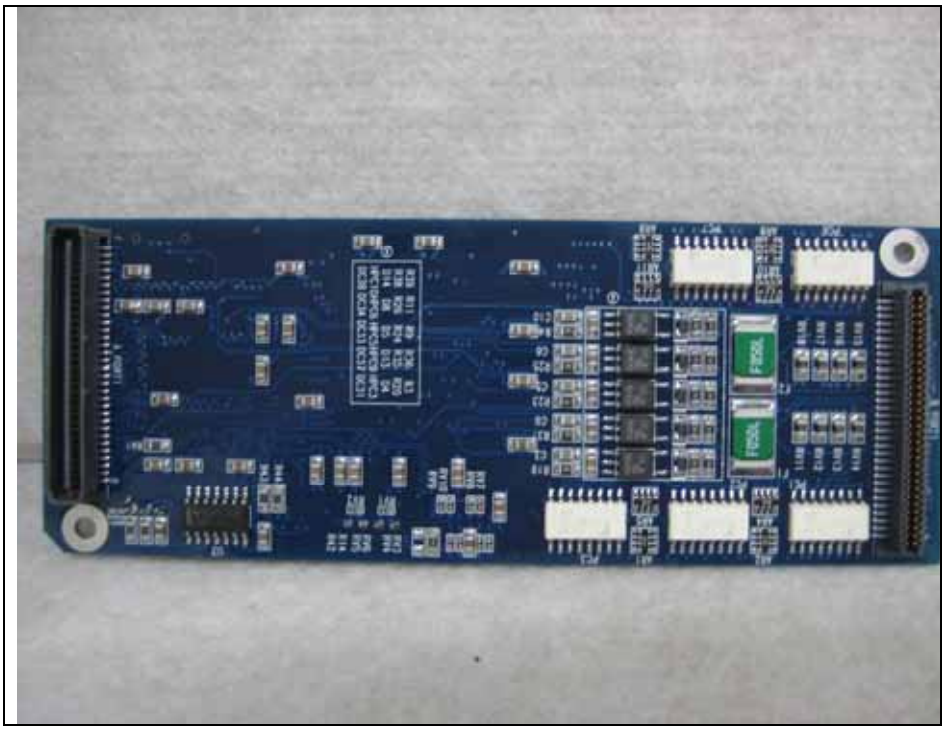




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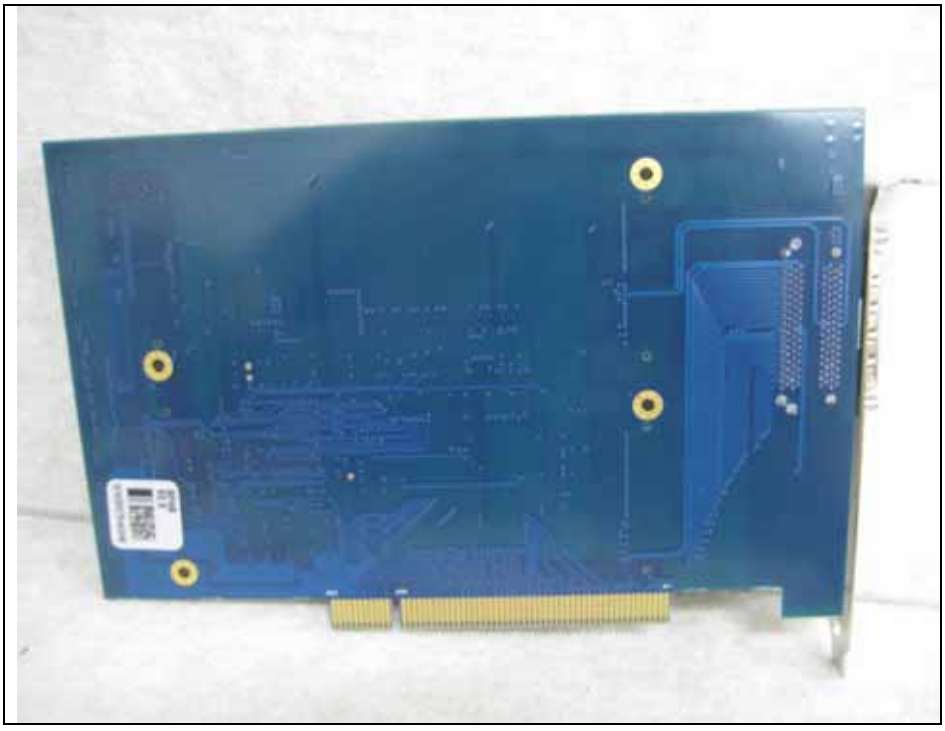
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-BPHR

Front View



Rear View



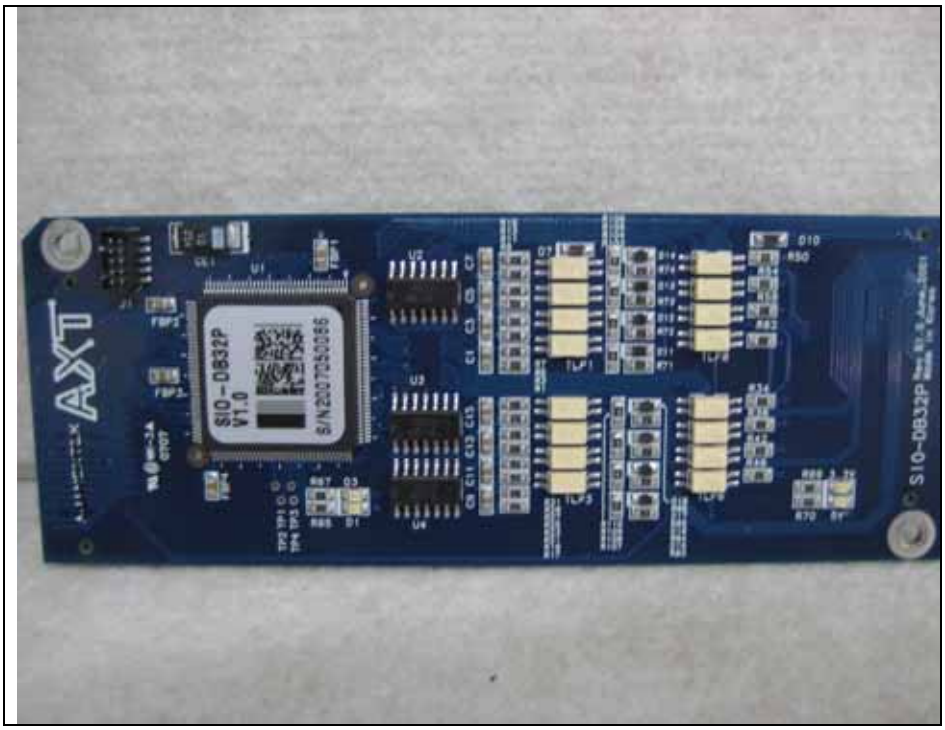
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Inside



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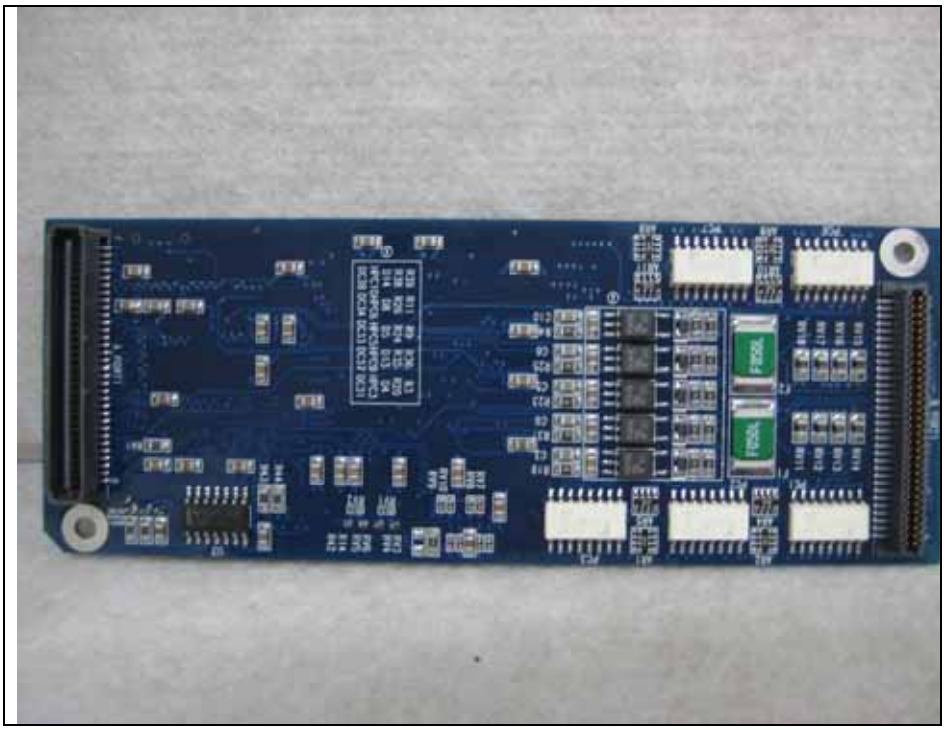




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